

STAT lecture 6

Confidence intervals (CI)

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Point estimation – estimation of unknown distribution parameter(s)

Random variable X , unknown parameter(s) of assumed distribution, θ

Random sample $X_1, \dots, X_n$ (iid)

Estimate $\hat{\theta} = h(x_1, \dots, x_n)$

gives a value when the data values of random sample are put in h function

Confidence interval (CI) for unknown distribution parameter(s)

CI = interval around the estimate $\hat{\theta}$ that with a probability, i.e. specified level of confidence, equal to $100(1 - \alpha)\%$ contains the unknown parameter θ

Usually α is chosen to be either 0.01, 0.05 or 0.10 which gives level of confidence for the CI as 99%, 95% and 90%, respectively

The bigger the confidence level is specified, the wider the CI will be.

Therefore it will be a compromise, CI's are desired to be narrow but then the cost will be that the level of confidence will be small(er)

Confidence interval for μ in normal distribution with σ^2 known

Model $X_1, \dots, X_n$ iid $X_i \sim N(\mu, \sigma^2)$, μ unknown, σ^2 known

Estimate $\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
 $Z \stackrel{\text{def}}{=} \frac{\hat{\mu} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$
 $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$

2-sided $100(1 - \alpha)\%$ CI for μ : $\left[\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$

1-sided lower bound $100(1 - \alpha)\%$ CI for μ : $\left[\bar{X} - z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty \right[$

1-sided upper bound $100(1 - \alpha)\%$ CI for μ : $] -\infty, \bar{X} + z_{\alpha} \frac{\sigma}{\sqrt{n}}]$

Confidence interval for μ in normal distribution with σ^2 unknown

Model $X_1, \dots, X_n$ iid $X_i \sim N(\mu, \sigma^2)$, μ unknown, σ^2 unknown

Estimates $\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
 $\hat{\sigma}^2 = s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \frac{\sigma^2}{n-1} \chi^2(n-1)$
 $T \stackrel{\text{def}}{=} \frac{\hat{\mu} - \mu}{\frac{s}{\sqrt{n}}} \sim t(n-1)$
 $P(-t(n-1)_{\alpha/2} \leq T \leq t(n-1)_{\alpha/2}) = 1 - \alpha$

2-sided $100(1 - \alpha)\%$ CI for μ : $\left[\bar{X} \pm t(n-1)_{\alpha/2} \frac{s}{\sqrt{n}} \right]$

1-sided lower bound $100(1 - \alpha)\%$ CI for μ : $\left[\bar{X} - t(n-1)_{\alpha} \frac{s}{\sqrt{n}}, \infty \right]$

1-sided upper bound $100(1 - \alpha)\%$ CI for μ : $\left] -\infty, \bar{X} + t(n-1)_{\alpha} \frac{s}{\sqrt{n}} \right]$

Confidence interval for σ^2 in normal distribution

Model $X_1, \dots, X_n$ iid $X_i \sim N(\mu, \sigma^2)$, μ unknown, σ^2 unknown

Estimate $\hat{\sigma}^2 = s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \frac{\sigma^2}{n-1} \chi^2(n-1)$

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

$$P\left(\chi^2(n-1)_{1-\alpha/2} \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi^2(n-1)_{\alpha/2}\right) = 1 - \alpha$$

2-sided $100(1 - \alpha)\%$ CI for σ^2 : $\left[\frac{(n-1)s^2}{\chi^2(n-1)_{\alpha/2}}, \frac{(n-1)s^2}{\chi^2(n-1)_{1-\alpha/2}} \right]$

Confidence interval for p in binomial distribution

Model $X \sim \text{Bin}(n, p)$

Estimate $\hat{p} = \frac{x}{n} \sim \frac{1}{n} \text{Bin}(n, p)$
 $E[\hat{p}] = p$
 $V[\hat{p}] = \frac{p(1-p)}{n}$

Due to Central Limit Theorem, for $np \gg 1$, $n(1-p) \gg 1$

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right) \quad (\text{approximately})$$
$$\mathbf{Z} \stackrel{\text{def}}{=} \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0, 1) \quad (\text{approximately})$$

2-sided $100(1 - \alpha)\%$ CI for p : $\left[\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right]$

Confidence (prediction) interval for next observation in normal distribution

Model $X_1, \dots, X_n$ iid $X_i \sim N(\mu, \sigma^2)$, μ unknown, σ^2 known

Next obs $X_{n+1} \sim N(\mu, \sigma^2)$

Estimate $\hat{X}_{n+1} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
 $\hat{X}_{n+1} - X_{n+1} \sim N\left(0, \sigma^2 \left(1 + \frac{1}{n}\right)\right)$

$$T \stackrel{\text{def}}{=} \frac{\hat{X}_{n+1} - X_{n+1}}{\sqrt{s^2 \left(1 + \frac{1}{n}\right)}} \sim t(n-1)$$

2-sided $100(1 - \alpha)\%$ CI for X_{n+1} : $\left[\hat{X}_{n+1} \pm t(n-1)_{\alpha/2} \sqrt{s^2 \left(1 + \frac{1}{n}\right)} \right]$